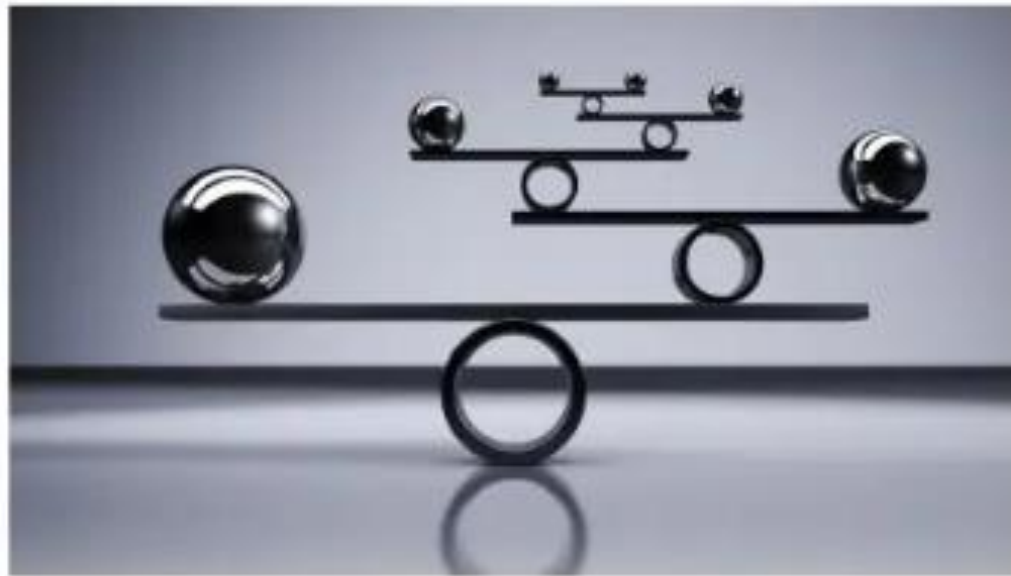


Software Construction and Development SE-312

LECTURE # 06

Lehman's Law

Course Instructor: Engr. Ruqqiya Asad



The Evolution of Software

Lehman's Law

Who is Lehman?

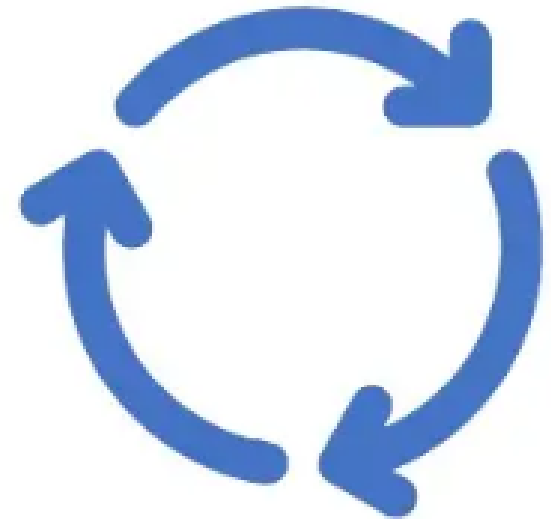
Lehman (1925, 2015) was a British computer scientist and professor of computing science at Imperial College London.

In the 1970s, he conducted studies on software evolution, leading to the formulation of his Laws of Software Evolution. These laws describe the inherent properties of software systems and how they change over time.

What is software evolution?

Software evolution is the ongoing process of changing and improving software throughout its life.

This includes all the updates, fixes, and enhancements made to the software after its initial development and release.

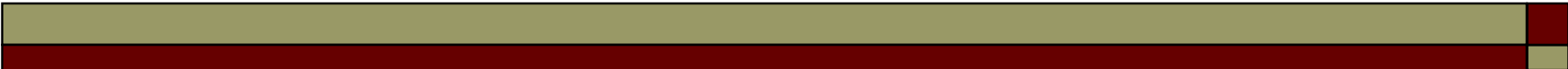


Lehman's Software Evolution laws

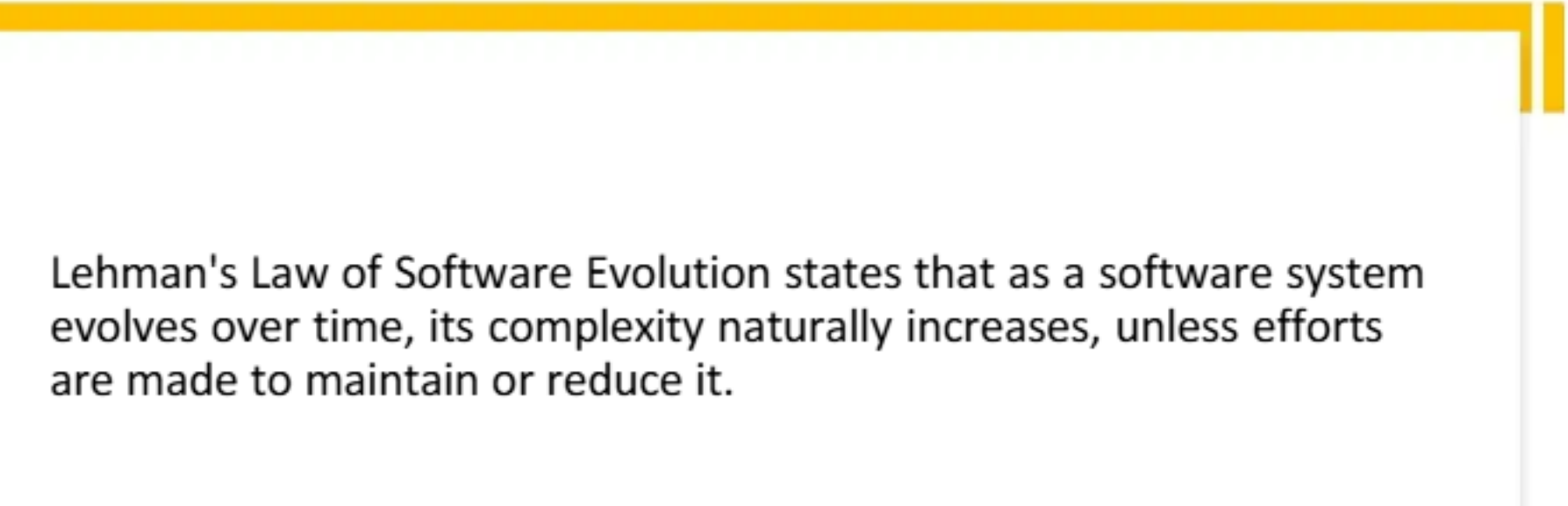
Lehman categorized software evolution into three types:

- Static Type
- Practical Type
- Embedded Type

These categories represent different aspects of how software evolves over time.



Lehman's Law



Lehman's Law of Software Evolution states that as a software system evolves over time, its complexity naturally increases, unless efforts are made to maintain or reduce it.

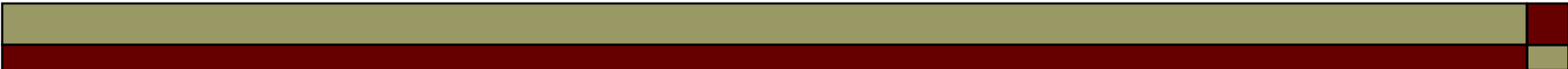
Key Principles of Lehman's Law

- 1: Continuing Change
- 2: Increasing Complexity
- 3: Self-regulation
- 4: Conservation of organizational stability
- 5: Conservation of familiarity
- 6: Continuous growth
- 7: Declining quality
- 8: Feedback system
- 9: Invisibility
- 10: Conservation of change

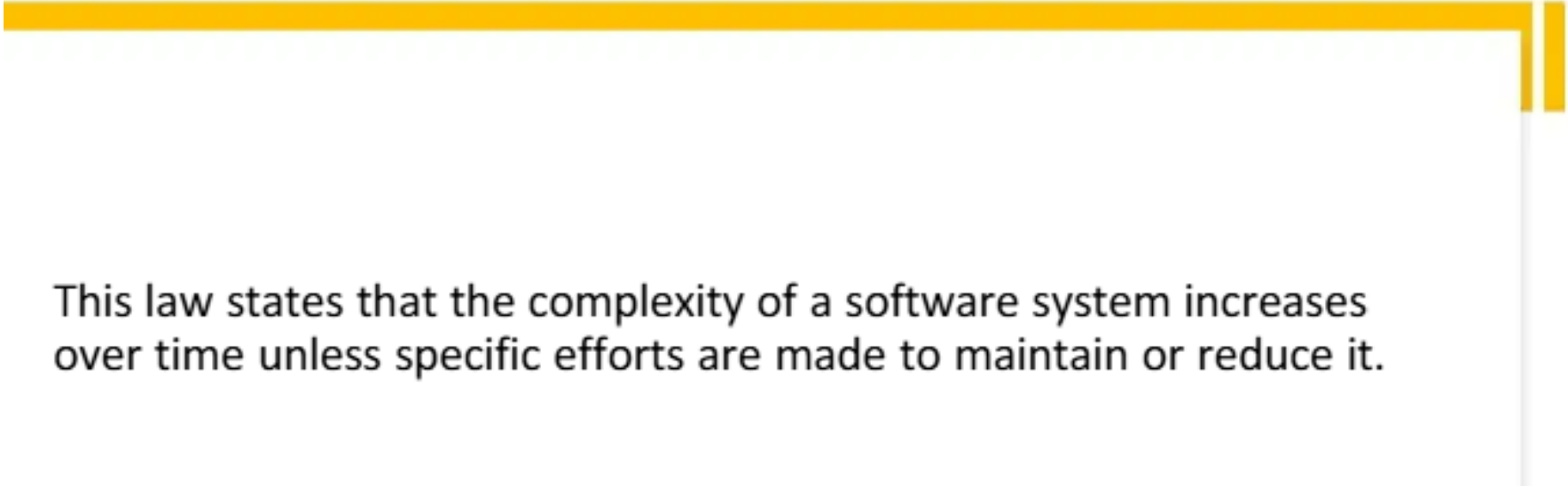
Principle 1: Continuing Change



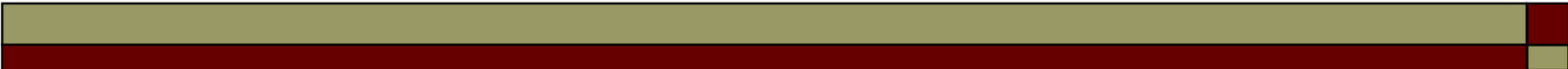
This law states that any software system representing real-world reality must undergo continuous change, or it will become less useful in that environment over time.



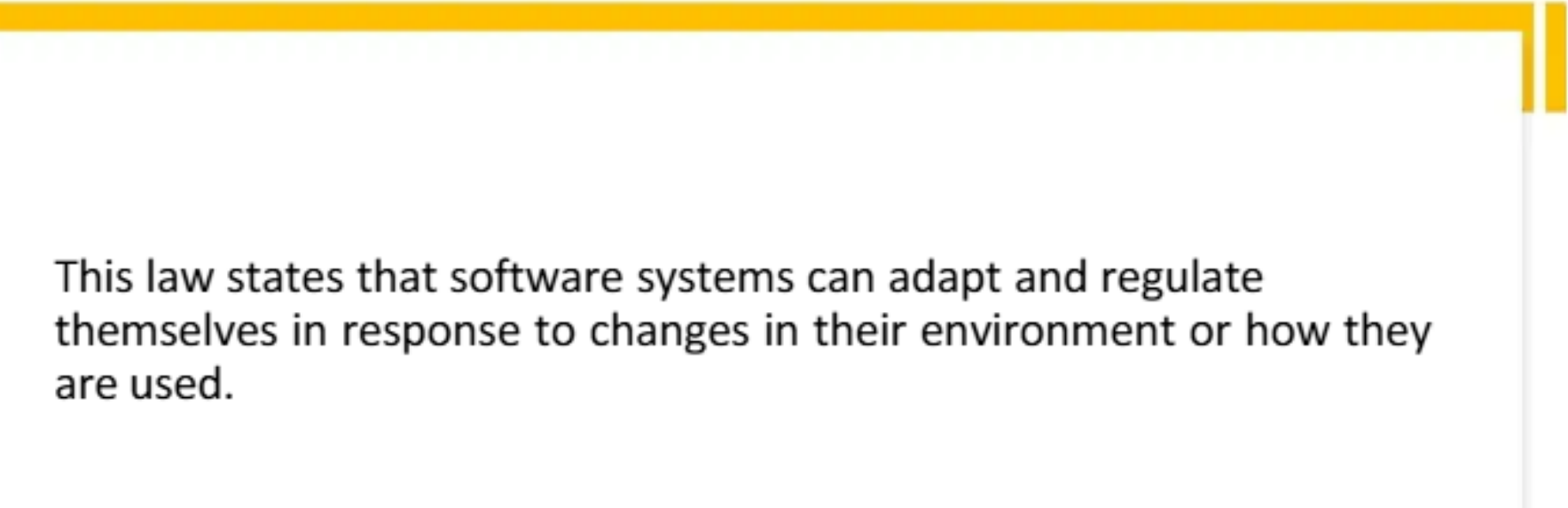
Principle 2: Increasing Complexity



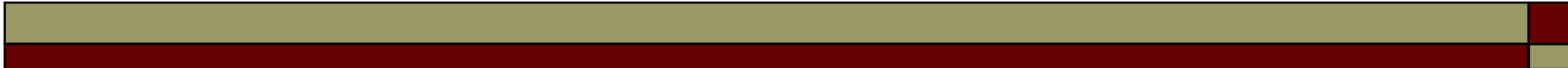
This law states that the complexity of a software system increases over time unless specific efforts are made to maintain or reduce it.



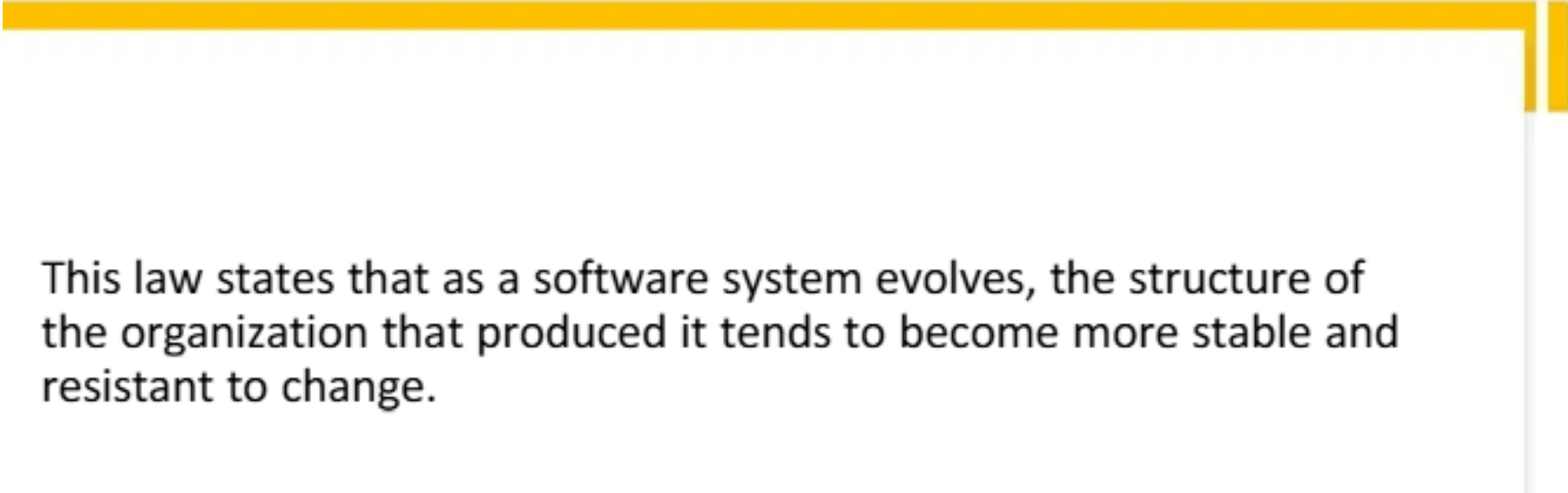
Principle 3: Self-Regulation



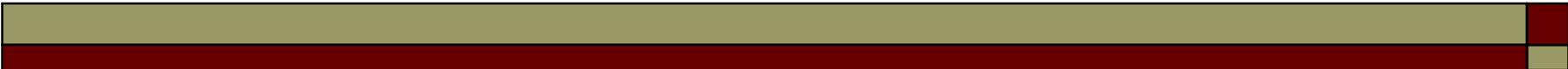
This law states that software systems can adapt and regulate themselves in response to changes in their environment or how they are used.



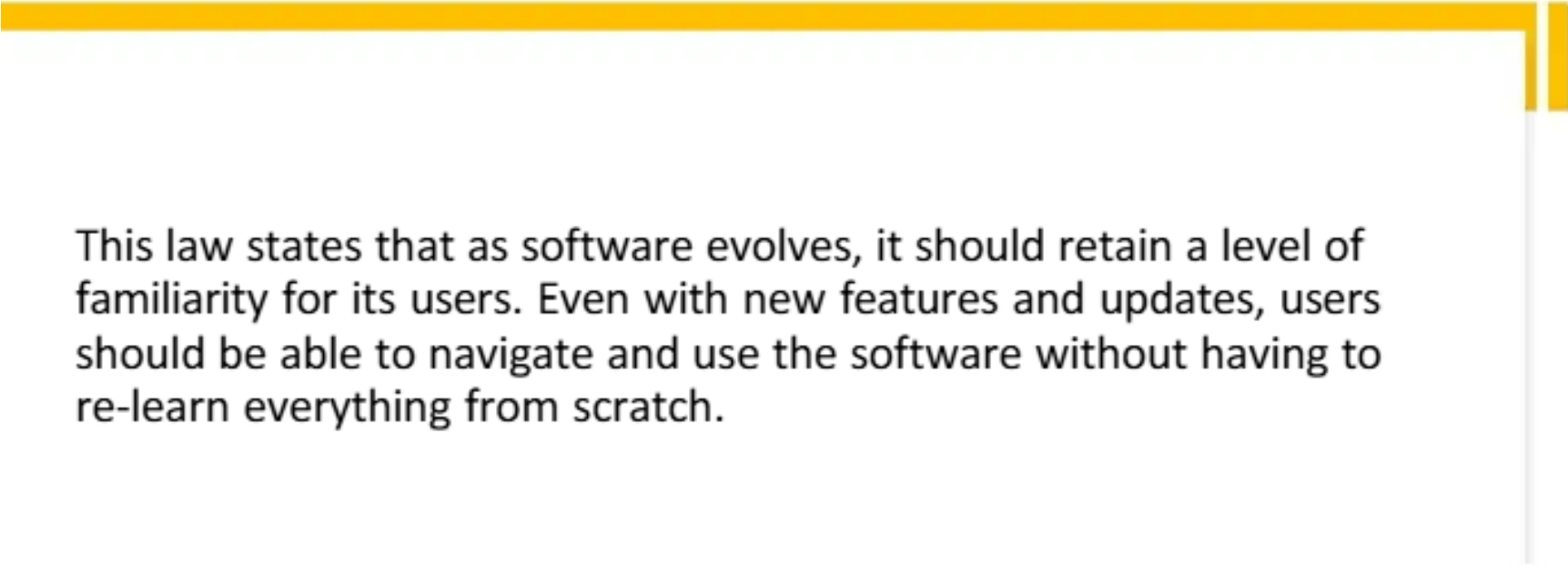
Principle 4: Conservation of Organizational Stability



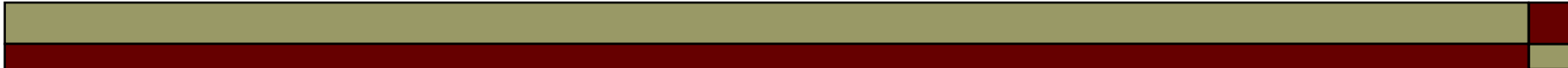
This law states that as a software system evolves, the structure of the organization that produced it tends to become more stable and resistant to change.



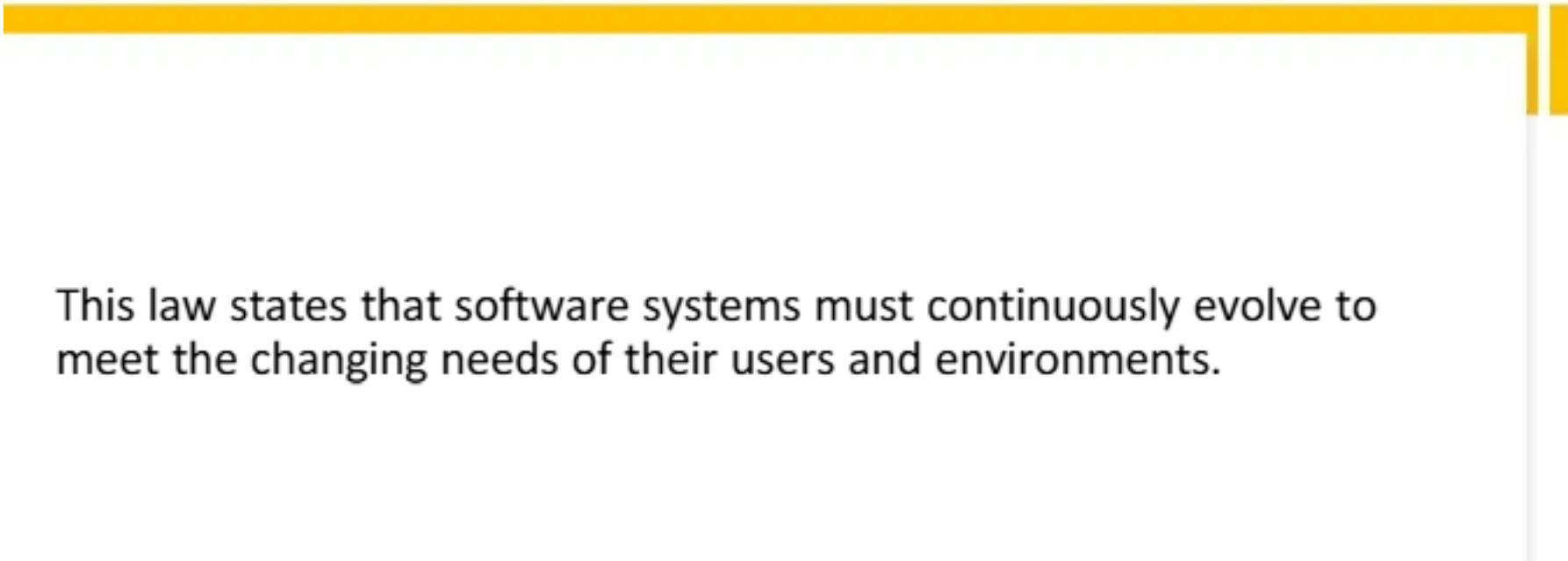
Principle 5: Conservation of Familiarity



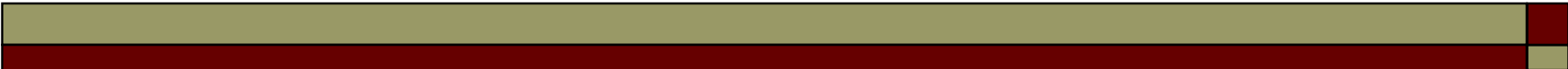
This law states that as software evolves, it should retain a level of familiarity for its users. Even with new features and updates, users should be able to navigate and use the software without having to re-learn everything from scratch.



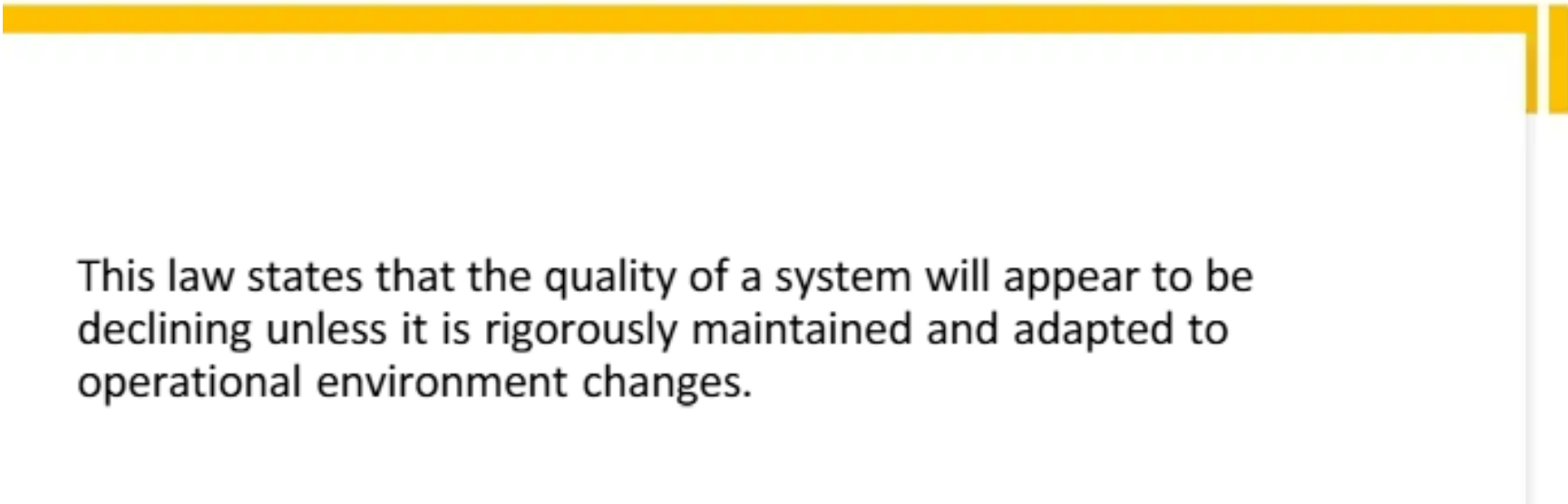
Principle 6: Continuous Growth



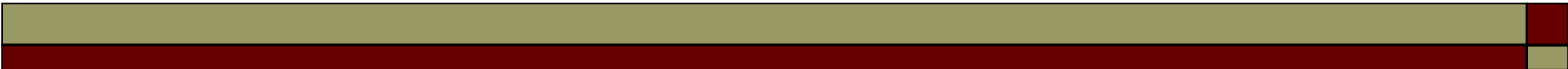
This law states that software systems must continuously evolve to meet the changing needs of their users and environments.



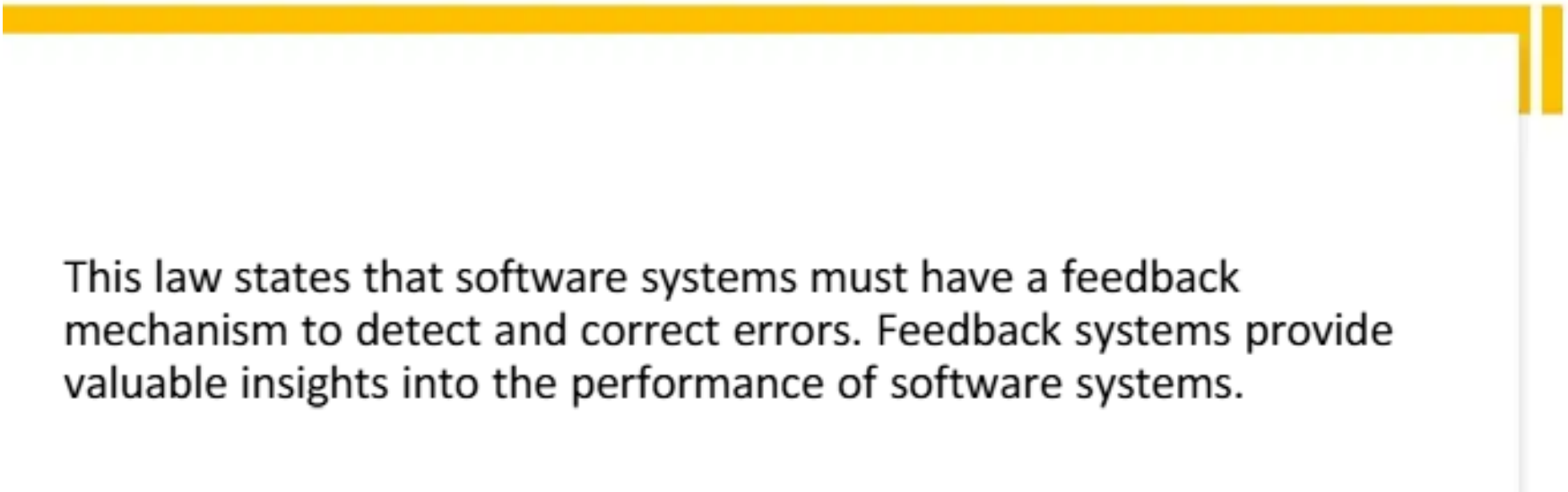
Principle 7: Declining Quality



This law states that the quality of a system will appear to be declining unless it is rigorously maintained and adapted to operational environment changes.

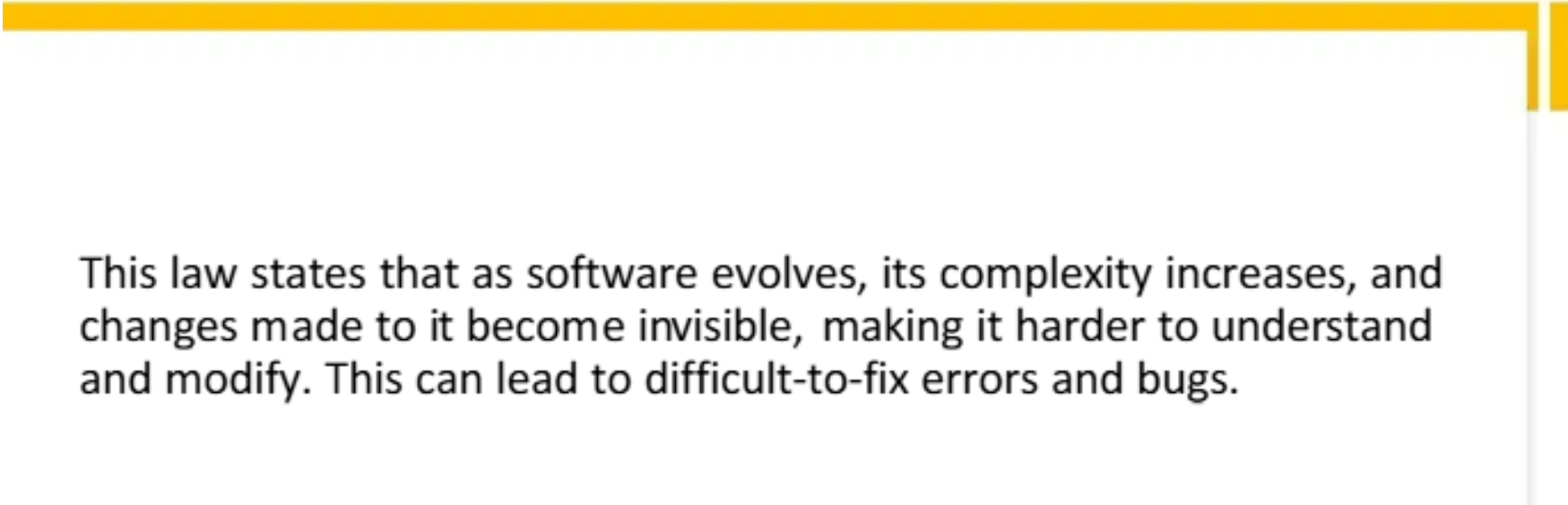


Principle 8: Feedback System



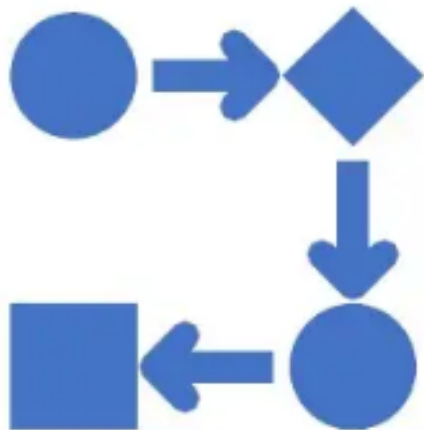
This law states that software systems must have a feedback mechanism to detect and correct errors. Feedback systems provide valuable insights into the performance of software systems.

Principle 9: Invisibility



This law states that as software evolves, its complexity increases, and changes made to it become invisible, making it harder to understand and modify. This can lead to difficult-to-fix errors and bugs.

Principle 10: Conservation of Change



- This law states that as a software system evolves, the rate of change decreases over time, but the amount of change remains constant.

Advantages of Implementing Lehman's Laws

Improved Software Quality:

- Implementing Lehman's Laws significantly improves software quality and reliability by focusing on maintaining software system stability for long-term success.

Reduced Maintenance Costs

- Adhering to Lehman's Laws can result in reduced maintenance costs for software development and evolution.

Advantages of Lehman's Law

- **Increased Productivity:**
 - Lehman's Laws increase software development team productivity by helping them understand and plan for the constant changes and complexity of software evolution.

Disadvantages of Not Implementing Lehman's law

- Reduced Software Quality
 - Neglecting Lehman's Laws can lead to reduced software quality and reliability. These laws help understand software evolution over time, and ignoring them can result in suboptimal outcomes.
- Increased Maintenance Cost
 - Ignoring Lehman's Laws can raise maintenance costs in software development. As software systems grow more complex, maintaining them becomes expensive without following Lehman's principles.

Disadvantages of Not Implementing Lehman's law

- Decreased Productivity:
 - Neglecting Lehman's Laws reduces software development team productivity. Evolving software systems become more complex and challenging to maintain over time.

Application of Lehman's Law



- SOFTWARE MAINTENANCE
- SOFTWARE UPGRADES AND PATCHES
- USER FEEDBACK AND USER EXPERIENCE
- TESTING AND QUALITY ASSURANCE
- FEATURE ENHANCEMENTS

Conclusion

- In conclusion, Lehman's Law of Software Evolution is a crucial aspect of software development that should not be overlooked. Understanding these principles can help software developers and organizations effectively address the challenges of software evolution and reap its benefits. By continuously adapting and maintaining software, developers can ensure its long-term usefulness and success in a changing environment.